Option Pricing: Motivation and First Mathematical Tools

Lecture 1 Notes — Class Support and Student Summary

1 Motivation, Replication, and Pricing

1.1 Contracts, payoffs, and discounting

A *financial contract* specifies future cash flows. A derivative links those cash flows to an observable market quantity. For a European call with strike K and maturity T,

$$X = (S_T - K)^+, \qquad (x)^+ = \max(x, 0).$$

The long position receives X at T and the short position owes X.

If the continuously compounded rate is constant and equal to r, the bank account is $B_t = e^{rt}$. A certain payment C at time T has present value Ce^{-rT}.

1.2 The casino example: expectation is not a price

Example — historical expectation versus replication

A contract pays \$100 after red and \$0 after black. Historically, $\mathbb{P}(\text{red}) = 0.99$, so its physical expectation is \$99. A \$50 casino bet pays \$100 after red and \$0 after black. It reproduces the contract state by state. Therefore the replication cost is \$50, not \$99.

Proposition 1.1 (Law of one price)**.** If two portfolios have the same payoff in every state, then they must have the same price in an arbitrage-free market.

Proof. If portfolio A were more expensive than B, sell A and buy B. The initial cash flow is positive and the terminal cash flow is zero in every state. This is an arbitrage. Interchange A and B for the opposite price inequality. □

The casino example isolates the central principle:

$$\boxed{\text{price of a replicable claim} = \text{cost of replication.}}$$

Historical probabilities describe frequencies; they do not override a state-by-state replication argument.

1.3 A one-period financial market

Let a stock satisfy $S_0 > 0$ and

$$S_1 = \begin{cases} S_u, & \text{up}, \\ S_d, & \text{down}, \end{cases} \qquad S_u > S_d,$$

and let the risk-free gross return be $R > 0$. A claim pays H_u or H_d.

Proposition 1.2 (Replication and the unique one-period price)**.** The portfolio with Δ shares and maturity cash B_1 replicates H when

$$\Delta = \frac{H_u - H_d}{S_u - S_d}, \qquad B_1 = H_d - \Delta S_d.$$

Its no-arbitrage price is

$$V_0 = \Delta S_0 + \frac{B_1}{R}.$$

Proof. The two replication equations are $\Delta S_u + B_1 = H_u$ and $\Delta S_d + B_1 = H_d$. Subtract them to obtain Δ, then substitute to obtain B_1. The law of one price identifies V_0 with the initial cost of this portfolio. □

Define

$$q = \frac{RS_0 - S_d}{S_u - S_d}.$$

The no-arbitrage inequalities $S_d < RS_0 < S_u$ are equivalent to $0 < q < 1$. Moreover,

$$S_0 = \frac{1}{R}\big(qS_u + (1-q)S_d\big), \qquad V_0 = \frac{1}{R}\big(qH_u + (1-q)H_d\big).$$

Thus q is the *risk-neutral up probability*. It is determined by traded prices, not by the physical frequency of the up state.

Example — replicate a call

Assume $R = 1$, $S_0 = 50$, $S_u = 60$, $S_d = 40$, and $K = 50$. The call pays $(10, 0)$. Hence

$$\Delta = \frac{10-0}{60-40} = \frac{1}{2}, \qquad B_1 = 0 - \frac{1}{2}(40) = -20, \qquad V_0 = \frac{1}{2}(50) - 20 = 5.$$

Also $q = (50-40)/(60-40) = 1/2$, so $V_0 = \mathbb{E}^q[H] = \frac{1}{2}(10) + \frac{1}{2}(0) = 5$.

Exercise — a similar one-period call

Assume $R = 1$, $S_0 = 100$, $S_u = 125$, $S_d = 75$, and $K = 100$. Find the replicating portfolio, price, and risk-neutral probability.

Solution

The call payoff is 25 in the up state and 0 in the down state. Therefore the replicating portfolio must satisfy

$$125\Delta + B_1 = 25, \qquad 75\Delta + B_1 = 0.$$

Subtracting the second equation from the first gives

$$50\Delta = 25, \qquad \Delta = \frac{1}{2}.$$

Substitution into the down-state equation gives $B_1 = -75/2 = -37.5$. Since $R = 1$, the time-zero cost is

$$V_0 = \Delta S_0 + \frac{B_1}{R} = \frac{1}{2}(100) - 37.5 = 12.5.$$

Finally, the risk-neutral probability is determined by the stock price:

$$100 = q(125) + (1-q)75, \qquad q = \frac{1}{2}.$$

This gives the same claim value, $V_0 = \mathbb{E}^q[H] = \frac{1}{2}(25) + \frac{1}{2}(0) = 12.5$.

2 Probability Toolbox

2.1 Random variables, distributions, CDFs, and densities

Definition 2.1 (Random variable and distribution)**.** A random variable is a measurable map $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its distribution is the probability measure

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \qquad A \in \mathcal{B}(\mathbb{R}).$$

Definition 2.2 (CDF and density)**.** The cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \le x).$$

It is nondecreasing, right-continuous, and satisfies $F_X(-\infty) = 0$, $F_X(+\infty) = 1$. If a nonnegative function f_X satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\,\mathrm{d}u,$$

then f_X is a density of X. At differentiability points, $f_X = F_X'$.

For $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

A distribution always exists; a density need not exist.

Example — a payoff is a random variable

Suppose

S_T	80	100	130
$\mathbb{P}$	0.20	0.50	0.30

and $X = (S_T - 100)^+$. Then X takes values $0, 0, 30$, so

$$\mathbb{P}(X = 0) = 0.70, \qquad \mathbb{P}(X = 30) = 0.30.$$

The payoff rule maps states into numbers; the probabilities of those numbers form its distribution.

Example — recover the exponential CDF

If $X \sim \mathrm{Exp}(\lambda)$, then $f_X(x) = \lambda e^{-\lambda x}\mathbf{1}_{\{x\ge 0\}}$. Therefore

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \ge 0. \end{cases}$$

Consequently, for $0 \le a \le b$, $\mathbb{P}(a < X \le b) = e^{-\lambda a} - e^{-\lambda b}$.

Exercise — CDF of a one-day return

Let $\mathbb{P}(R = -2\%) = 0.40$ and $\mathbb{P}(R = 3\%) = 0.60$. Write F_R and compute $\mathbb{P}(R > 0)$.

Solution

The CDF accumulates the probability mass at and below the point r. Before -2% no value is possible; at -2% it jumps by 0.40, and at 3% it jumps by the remaining 0.60. Hence

$$F_R(r) = \begin{cases} 0, & r < -2\%, \\ 0.40, & -2\% \le r < 3\%, \\ 1, & r \ge 3\%, \end{cases}.$$

Since 0 lies between the two possible returns, $F_R(0) = 0.40$. Therefore

$$\mathbb{P}(R > 0) = 1 - F_R(0) = 1 - 0.40 = 0.60.$$

Equivalently, the only positive outcome is $R = 3\%$, which has probability 0.60.

2.2 Expectation, variance, and dependence

Definition 2.3 (Expectation and moments)**.** For integrable X,

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x)\,\mathrm{d}x.$$

More generally, whenever it exists,

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f_X(x)\,\mathrm{d}x.$$

The kth moment is $\mathbb{E}[X^k]$. In particular,

$$\mu = \mathbb{E}[X], \quad m_2 = \mathbb{E}[X^2], \quad \operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2], \quad \operatorname{SD}(X) = \sqrt{\operatorname{Var}(X)}.$$

Proposition 2.4 (Fundamental expectation properties)**.** For integrable X, Y and constants a, b,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y], \qquad X \le Y \Longrightarrow \mathbb{E}[X] \le \mathbb{E}[Y], \qquad |\mathbb{E}[X]| \le \mathbb{E}[|X|].$$

Also $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

Proof. Linearity follows from linearity of sums or integrals. If $X \le Y$, then $Y - X \ge 0$ and $\mathbb{E}[Y - X] \ge 0$. Finally, $-|X| \le X \le |X|$ and monotonicity give the absolute-value inequality. The indicator identity follows directly from its two possible values. □

Proposition 2.5 (Variance and covariance identities)**.**

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

With $\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$,

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y).$$

If X and Y are independent and square integrable, then $\operatorname{Cov}(X, Y) = 0$.

Proof. Expand $(X-\mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations. Expanding the square of $a(X - \mathbb{E}[X]) + b(Y - \mathbb{E}[Y])$ gives the second identity. Independence implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, hence zero covariance. □

Law	Parameters	$\mathbb{E}[X]$	$\operatorname{Var}(X)$
Bernoulli	p	p	$p(1-p)$
Binomial	n, p	np	$np(1-p)$
Poisson	λ	λ	λ
Uniform	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential	rate λ	$1/\lambda$	$1/\lambda^2$
Gaussian	μ, σ^2	μ	σ^2

Example — moments of a Bernoulli sum

Let $B_1, \dots, B_n$ be independent Bernoulli(p) variables and $N_n = \sum_{k=1}^{n} B_k$. Since $B_k^2 = B_k$,

$$\mathbb{E}[B_k] = p, \qquad \operatorname{Var}(B_k) = p - p^2 = p(1-p).$$

Linearity gives $\mathbb{E}[N_n] = np$. Independence removes cross-covariances, so $\operatorname{Var}(N_n) = np(1-p)$.

Exercise — classical moment calculations

Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\operatorname{Var}(X)$ for a uniform and an exponential variable. Identify the odd moments of a standard Gaussian. As an extension, derive the Poisson variance from $\mathbb{E}[X(X-1)]$.

Solution

For $X \sim \operatorname{Unif}[a, b]$, the density is $1/(b-a)$ on $[a, b]$. Thus

$$\mathbb{E}[X] = \frac{1}{b-a}\int_a^b x\,\mathrm{d}x = \frac{a+b}{2}, \qquad \mathbb{E}[X^2] = \frac{1}{b-a}\int_a^b x^2\,\mathrm{d}x = \frac{a^2+ab+b^2}{3}.$$

Consequently,

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(b-a)^2}{12}.$$

For $X \sim \text{Exp}(\lambda)$, integration by parts (or the gamma integral) gives

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\,\mathrm{d}x = \frac{1}{\lambda}, \qquad \mathbb{E}[X^2] = \int_0^\infty x^2\lambda e^{-\lambda x}\,\mathrm{d}x = \frac{2}{\lambda^2}.$$

Hence $\text{Var}(X) = 2/\lambda^2 - (1/\lambda)^2 = 1/\lambda^2$. In summary,

X	$\mathbb{E}[X]$	$\mathbb{E}[X^2]$	$\text{Var}(X)$
$\text{Unif}[a,b]$	$(a+b)/2$	$(a^2+ab+b^2)/3$	$(b-a)^2/12$
$\text{Exp}(\lambda)$	$1/\lambda$	$2/\lambda^2$	$1/\lambda^2$

For $Z \sim \mathcal{N}(0,1)$, the density is even while z^{2m+1} is odd, so every odd moment vanishes: $\mathbb{E}[Z^{2m+1}] = 0$. Finally, if $N \sim \text{Poisson}(\lambda)$, then

$$\begin{aligned}\mathbb{E}[N(N-1)] &= e^{-\lambda}\sum_{k=2}^{\infty} k(k-1)\frac{\lambda^k}{k!} \\ &= \lambda^2 e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda^2.\end{aligned}$$

Since $N^2 = N(N-1) + N$ and $\mathbb{E}[N] = \lambda$, we obtain $\mathbb{E}[N^2] = \lambda^2 + \lambda$ and therefore $\text{Var}(N) = \lambda$.

2.3 Sample averages, LLN, and CLT

For observations $X_1, \ldots, X_n$, define

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i, \qquad v_n = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X}_n)^2, \qquad s_n^2 = \frac{n}{n-1}v_n.$$

v_n is the empirical variance; s_n^2 is its unbiased version under i.i.d. sampling with finite variance.

Theorem 2.6 (Law of Large Numbers)**.** If $X_1, X_2, \ldots$ are i.i.d. and $\mathbb{E}[|X_1|] < \infty$, then

$$\overline{X}_n \longrightarrow \mu = \mathbb{E}[X_1] \qquad \text{in probability.}$$

Easy proof under the stronger assumption $\text{Var}(X_1) = \sigma^2 < \infty$. Independence gives $\mathbb{E}[\overline{X}_n] = \mu$ and $\text{Var}(\overline{X}_n) = \sigma^2/n$. By Chebyshev's inequality,

$$\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

□

Theorem 2.7 (Central Limit Theorem)**.** If, in addition, $0 < \sigma^2 = \text{Var}(X_1) < \infty$, then

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0,1).$$

The LLN says the estimation error vanishes. The CLT says its typical size is $\sigma/\sqrt{n}$ and gives an approximate Gaussian shape. This distinction is fundamental for statistics, simulation error, and averages of financial data.

Example — frequency of successes

For independent Bernoulli(0.60) variables,

$$\mathbb{E}[\overline{X}_{100}] = 0.60, \qquad \text{SD}(\overline{X}_{100}) = \sqrt{\frac{0.60(0.40)}{100}} \approx 0.049.$$

The CLT gives

$$\mathbb{P}(0.50 \le \overline{X}_{100} \le 0.70) \approx \mathbb{P}(-2.04 \le Z \le 2.04) \approx 0.96.$$

Exercise — average daily loss

Let $X_1, \dots, X_{225}$ be i.i.d. with mean 2 and standard deviation 3. Find the LLN limit, the standard deviation of $\overline{X}_{225}$, and a CLT approximation of $\mathbb{P}(\overline{X}_{225} > 2.4)$.

Solution

The variables are i.i.d. with finite mean, so the LLN gives

$$\overline{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} 2.$$

In particular, $\overline{X}_{225}$ is centered at $\mathbb{E}[\overline{X}_{225}] = 2$ and should be close to 2 with high probability. Independence also gives

$$\mathrm{Var}(\overline{X}_{225}) = \frac{\mathrm{Var}(X_1)}{225} = \frac{9}{225} = 0.04, \qquad \mathrm{SD}(\overline{X}_{225}) = 0.2.$$

For the tail probability, standardize the threshold using this standard deviation. The CLT yields

$$\mathbb{P}(\overline{X}_{225} > 2.4) \approx \mathbb{P}\left(Z > \frac{2.4 - 2}{0.2}\right) = \mathbb{P}(Z > 2) = 1 - \Phi(2) \approx 0.0228.$$

2.4 Characteristic functions and joint distributions

Definition 2.8 (Characteristic function)**.**

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \qquad u \in \mathbb{R}.$$

It always exists, satisfies $|\varphi_X(u)| \le 1$, and uniquely determines the distribution of X.

Proposition 2.9 (Independent sums become products)**.** If X and Y are independent, then

$$\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u).$$

Proof. Independence gives

$$\mathbb{E}[e^{iu(X+Y)}] = \mathbb{E}[e^{iuX}e^{iuY}] = \mathbb{E}[e^{iuX}]\mathbb{E}[e^{iuY}].$$

□

A joint distribution records probabilities involving several random variables. Marginals are obtained by summing or integrating out the other variables. The joint law, unlike the two marginals alone, records dependence.

3 Information and Conditional Expectation

3.1 Signals, sigma-algebras, and measurability

Example — a signal reveals a partition

Roll a fair die and let $X(\omega) = \omega$. Observe only the parity signal $Y = \mathbf{1}_{\{X \text{ is even}\}}$. The observation distinguishes

$$O = \{1, 3, 5\}, \qquad E = \{2, 4, 6\},$$

but cannot distinguish outcomes inside either set.

Definition 3.1 (Sigma-algebra and information generated by a signal)**.** A collection $\mathcal{G}$ of events is a sigma-algebra if $\Omega \in \mathcal{G}$, it is closed under complements, and it is closed under countable unions. For a random variable Y,

$$\sigma(Y) = \{Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

is the information revealed by observing Y.

For the parity signal, $\sigma(Y) = \{\varnothing, O, E, \Omega\}$.

Definition 3.2 (Measurable random variable)**.** A random variable Z is $\mathcal{G}$-measurable if

$$\{Z \in B\} \in \mathcal{G} \qquad \text{for every Borel set } B.$$

It is then completely determined by the information in $\mathcal{G}$.

Proposition 3.3 (Finite-partition test)**.** If $\mathcal{G}$ is generated by a finite partition, then Z is $\mathcal{G}$-measurable if and only if Z is constant on each atom of that partition. Equivalently,

$$Z \text{ is } \sigma(Y)\text{-measurable} \iff Z = g(Y) \text{ for some measurable } g.$$

Proof. If $Z = g(Y)$, equal values of Y imply equal values of Z. Conversely, if Z is constant on every level set of Y, define $g(y)$ to be that common value on $\{Y = y\}$. In a finite model this defines a measurable function g. □

Example — test measurability on parity atoms

For $Z_1 = 10 - \mathbf{1}_E$, $Z_2 = X$, and $Z_3 = (-1)^X$:

- Z_1 is measurable: it is constant on O and on E;
- Z_2 is not measurable: parity does not determine the die value;
- Z_3 is measurable: it is a function of parity.

Exercise — a coarse price signal

Let $S_T \in \{80, 100, 130\}$ and $Y = \mathbf{1}_{\{S_T \geq 100\}}$. List $\sigma(Y)$. Decide whether $(S_T - 100)^+$ and $\mathbf{1}_{\{S_T \geq 100\}}$ are $\sigma(Y)$-measurable.

Solution

The signal is 0 when $S_T = 80$ and 1 when $S_T \in \{100, 130\}$. Its two information atoms are therefore $\{80\}$ and $\{100, 130\}$, and

$$\sigma(Y) = \{\varnothing, \Omega, \{80\}, \{100, 130\}\}.$$

A random variable is $\sigma(Y)$-measurable precisely when it is constant on each atom. The call payoff $(S_T - 100)^+$ equals 0 at $S_T = 100$ and 30 at $S_T = 130$. It is not constant on the second atom, so it is not $\sigma(Y)$-measurable. By contrast, the digital payoff is 0 on $\{80\}$ and 1 on $\{100, 130\}$; it is exactly Y and is therefore measurable.

3.2 Conditional distributions and conditional expectation

For events with $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

Conditioning replaces the marginal distribution by the distribution relevant after the signal has been observed.

Definition 3.4 (Conditional expectation)**.** Let $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$. A conditional expectation of X given $\mathcal{G}$ is a $\mathcal{G}$-measurable random variable Z such that

$$\int_A Z \, d\mathbb{P} = \int_A X \, d\mathbb{P} \qquad \text{for every } A \in \mathcal{G}.$$

It is denoted by $\mathbb{E}[X \mid \mathcal{G}]$. Conditioning on Y means conditioning on $\sigma(Y)$: $\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)]$.

Proposition 3.5 (Uniqueness)**.** Conditional expectation is unique up to almost-sure equality.

Proof. If Z and Z' both satisfy the definition, then $\int_A (Z - Z') \, d\mathbb{P} = 0$ for all $A \in \mathcal{G}$. Since $Z - Z'$ is measurable, choose $A = \{Z > Z'\}$ and $A = \{Z < Z'\}$ to conclude $Z = Z'$ almost surely. □

Proposition 3.6 (Finite-partition formula)**.** If $\mathcal{G}$ is generated by atoms $A_1, \ldots, A_m$ with positive probabilities, then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{j=1}^{m} \frac{\mathbb{E}[X \mathbf{1}_{A_j}]}{\mathbb{P}(A_j)} \mathbf{1}_{A_j} = \sum_{j=1}^{m} \mathbb{E}[X \mid A_j] \mathbf{1}_{A_j}.$$

Proof. The proposed variable is constant on each atom, hence $\mathcal{G}$-measurable. Its integral over A_j equals $\mathbb{E}[X \mid A_j]\mathbb{P}(A_j) = \mathbb{E}[X \mathbf{1}_{A_j}]$. Add these identities for any union of atoms. □

Example — prediction from a partial signal

On four equally likely states, let $X = (0, 4, 8, 12)$ and let $Y = L$ on states $\{1, 2\}$ and $Y = H$ on states $\{3, 4\}$. Then

$$\mathbb{E}[X \mid Y] = 2\mathbf{1}_{\{Y=L\}} + 10\mathbf{1}_{\{Y=H\}}.$$

The conditional prediction is the average inside the observed information set.

Exercise — a payoff after a coarse die signal

Roll a fair die and let $X = 12\mathbf{1}_{\{\omega=6\}}$. Observe only $L = \{1, 2, 3, 4\}$ or $H = \{5, 6\}$. Compute $\mathbb{E}[X \mid \sigma(L, H)]$ and verify its expectation.

Solution

On L, the die cannot equal 6, so $X = 0$ throughout that atom and $\mathbb{E}[X \mid L] = 0$. On $H = \{5, 6\}$, the two outcomes remain equally likely and the corresponding payoffs are 0 and 12. Therefore

$$\mathbb{E}[X \mid H] = \frac{0 + 12}{2} = 6.$$

The conditional expectation is thus the random variable

$$\mathbb{E}[X \mid \sigma(L, H)] = 6\mathbf{1}_H.$$

To verify the tower property, note that

$$\mathbb{E}[\mathbb{E}[X \mid \sigma(L, H)]] = 6\mathbb{P}(H) = 6\frac{2}{6} = 2.$$

Directly, $X = 12$ only when the die equals 6, so $\mathbb{E}[X] = 12(1/6) = 2$ as well.

3.3 Conditioning with a joint density

Suppose (X, Y) has joint density $f_{X,Y}$. When $f_Y(y) > 0$,

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}x, \qquad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

For integrable h,

$$g(y) = \int_{\mathbb{R}} h(x) f_{X|Y}(x \mid y)\,\mathrm{d}x = \mathbb{E}[h(X) \mid Y = y], \qquad \mathbb{E}[h(X) \mid Y] = g(Y).$$

Example — the upper triangular density

Let $f_{X,Y}(x, y) = 2\mathbf{1}_{\{0<x<y<1\}}$. For $0 < y < 1$,

$$f_Y(y) = \int_0^y 2\,\mathrm{d}x = 2y, \qquad f_{X|Y}(x \mid y) = \frac{1}{y}\mathbf{1}_{(0,y)}(x).$$

Thus $X \mid (Y = y) \sim \mathrm{Unif}(0, y)$ and

$$\mathbb{E}[X \mid Y = y] = \frac{y}{2}, \qquad \mathbb{E}[X \mid Y] = \frac{Y}{2}.$$

Exercise — the reversed triangular density

Let $f_{X,Y}(x, y) = 2\mathbf{1}_{\{0<y<x<1\}}$. Find f_Y, $f_{X|Y}$, and $\mathbb{E}[X \mid Y]$.

Solution

Fix $y \in (0,1)$. The support condition $0 < y < x < 1$ allows x to range from y to 1. Integrating the joint density over this interval gives

$$f_Y(y) = \int_y^1 2\,\mathrm{d}x = 2(1-y).$$

After normalization,

$$f_{X|Y}(x \mid y) = \frac{2\mathbf{1}_{(y,1)}(x)}{2(1-y)} = \frac{1}{1-y}\mathbf{1}_{(y,1)}(x).$$

Thus $X \mid (Y = y) \sim \mathrm{Unif}(y, 1)$. Its conditional mean can be read from the midpoint of the interval or calculated directly:

$$\mathbb{E}[X \mid Y = y] = \frac{1}{1-y}\int_y^1 x\,\mathrm{d}x = \frac{1-y^2}{2(1-y)} = \frac{1+y}{2}.$$

Replacing y by the observed random value Y gives

$$\mathbb{E}[X \mid Y] = \frac{1+Y}{2}.$$

3.4 Gaussian conditioning

Proposition 3.7 (Conditional law of a Gaussian pair)**.** Suppose (X, Y) is jointly Gaussian, with

$$\mathbb{E}[X] = \mu_X, \quad \mathbb{E}[Y] = \mu_Y, \quad \mathrm{Var}(X) = \sigma_X^2, \quad \mathrm{Var}(Y) = \sigma_Y^2 > 0,$$

and $\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y$. Then

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y),\ \sigma_X^2(1-\rho^2)\right).$$

In particular,

$$\mathbb{E}[X \mid Y] = \mu_X + \rho\frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \qquad \mathrm{Var}(X \mid Y) = \sigma_X^2(1-\rho^2).$$

Short derivation. Set $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$ and $R = X - \mu_X - a(Y - \mu_Y)$. Then $\mathrm{Cov}(R, Y) = 0$. Since (R, Y) is jointly Gaussian, zero covariance implies independence. Writing $X = \mu_X + a(Y - \mu_Y) + R$ gives the conditional mean above, while $\mathrm{Var}(R) = \sigma_X^2(1-\rho^2)$ gives the conditional variance. □

The conditional mean is the best affine prediction from Y. The remaining variance decreases as $|\rho|$ increases and vanishes when $|\rho| = 1$.

3.5 Properties used in dynamic pricing

Proposition 3.8 (Core conditional-expectation properties)**.** For integrable variables and appropriate sigma-algebras:

$$\begin{aligned}
\mathbb{E}[aX + bY \mid \mathcal{G}] &= a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}],\\
X \le Y &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}],\\
Z\ \mathcal{G}\text{-measurable} &\Longrightarrow \mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}],\\
X\ \text{independent of}\ \mathcal{G} &\Longrightarrow \mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X],\\
\mathcal{H} \subseteq \mathcal{G} &\Longrightarrow \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}].
\end{aligned}$$

In particular, $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.

Proof guide. For linearity, check the defining integral on every $A \in \mathcal{G}$ and use uniqueness. Positivity follows by testing the set where the conditional expectation would be negative. For a simple known factor $Z = \sum z_j \mathbf{1}_{A_j}$, pull each constant z_j through the defining integrals; approximation gives the general case. If X is independent of $\mathcal{G}$, the constant $\mathbb{E}[X]$ has the required averages. For the tower rule, test both sides on every $A \in \mathcal{H} \subseteq \mathcal{G}$ and use uniqueness again. □

4 Processes, Filtrations, and Martingales

4.1 Information evolving through time

Definition 4.1 (Stochastic process)**.** A discrete-time stochastic process $(X_n)_{n\geq 0}$ is a sequence of random variables on the same probability space. For fixed n, X_n is a random variable; for fixed ω, $n \mapsto X_n(\omega)$ is a sample path.

Definition 4.2 (Filtration and adapted process)**.** A filtration is an increasing family of sigma-algebras

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots .$$

It represents information available through time. A process (X_n) is adapted if X_n is $\mathcal{F}_n$-measurable for every n.

Adapted means "no look-ahead": the value at time n can use the observed past and present, but not a future observation.

Exercise — detect look-ahead

Let $\xi_1, \xi_2 \in \{0, 1\}$ encode two coin tosses and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Consider

$$X_0 = 0, \qquad X_1 = \xi_1 + \xi_2, \qquad X_2 = \xi_1 + \xi_2.$$

Is (X_n) adapted? Give an adapted cumulative-heads process.

Solution

At time 1, the sigma-algebra $\mathcal{F}_1 = \sigma(\xi_1)$ reveals only the first toss. Two outcomes can therefore have the same value of ξ_1 but different values of ξ_2. On those outcomes, $X_1 = \xi_1 + \xi_2$ is different, so X_1 is not $\mathcal{F}_1$-measurable. The original process is consequently not adapted.
The cumulative number of observed heads is instead

$$\widetilde{X}_0 = 0, \qquad \widetilde{X}_1 = \xi_1, \qquad \widetilde{X}_2 = \xi_1 + \xi_2.$$

Here $\widetilde{X}_n$ depends only on tosses observed by time n, so it is $\mathcal{F}_n$-measurable for each n and the process is adapted.

4.2 Random walks and martingales

Let

$$S_n = S_0 + \sum_{k=1}^{n} \varepsilon_k,$$

where the increments are i.i.d. with mean m and variance σ^2.

Proposition 4.3 (Random-walk moments)**.**

$$\mathbb{E}[S_n] = S_0 + nm, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

Proof. Use linearity for the mean. For the variance, independence removes all cross-covariances, leaving the sum of n identical variances. □

Definition 4.4 (Martingale)**.** An integrable adapted process (M_n) is a martingale with respect to $(\mathcal{F}_n)$ if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for every } n.$$

Given all current information, the best prediction of tomorrow's value is today's value.

Example — the centered random walk is a martingale

Let $M_n = \sum_{k=1}^{n} \varepsilon_k$, with independent centered increments and $\mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n)$. Since M_n is known at time n and ε_{n+1} is independent of $\mathcal{F}_n$,

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[M_n + \varepsilon_{n+1} \mid \mathcal{F}_n] = M_n + \mathbb{E}[\varepsilon_{n+1}] = M_n.$$

Exercise — a second random-walk martingale

Let $S_n = \sum_{k=1}^{n} \varepsilon_k$ for independent symmetric ± 1 increments. Show that $M_n = S_n^2 - n$ is a martingale.

Solution

The variable $M_n = S_n^2 - n$ is integrable and $\mathcal{F}_n$-measurable because it is a function of the first n increments. It remains to verify the martingale condition. Since ε_{n+1} is independent of $\mathcal{F}_n$ and is symmetric,

$$\mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] = 1.$$

Using $S_{n+1} = S_n + \varepsilon_{n+1}$ and pulling the known quantity S_n out of the conditional expectation,

$$\begin{aligned} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[(S_n + \varepsilon_{n+1})^2 - (n+1) \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n \mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] - (n+1) \\ &= S_n^2 - n = M_n. \end{aligned}$$

Thus all three requirements—integrability, adaptation, and the conditional expectation identity—are satisfied.

Fair games and financial pricing

In a fair casino game, conditional expected future wealth equals current wealth. Under a risk-neutral measure $\mathbb{Q}$, discounted traded-asset prices are martingales. For a replicable payoff X,

$$\frac{V_t}{B_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{X}{B_T} \,\middle|\, \mathcal{F}_t\right].$$

Replication supplies the economic restriction; conditional expectation and martingales express it dynamically.

5 Short Differential-Calculus Review

Definition 5.1 (Derivative)**.** For $g : \mathbb{R} \to \mathbb{R}$, if the limit exists,

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}.$$

Equivalently, $g(x+h) = g(x) + g'(x)h + o(h)$.

Definition 5.2 (Partial derivatives, gradient, and Hessian)**.** For $f : \mathbb{R}^d \to \mathbb{R}$ and the ith coordinate vector e_i,

$$\partial_i f(x) = \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h}, \qquad \nabla f(x) = (\partial_1 f(x), \ldots, \partial_d f(x)).$$

If second partial derivatives exist, the Hessian is

$$\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x)\right)_{i,j}.$$

Example — gradient and Hessian

For $f(x, y) = x^2 y + e^y$,

$$f_x = 2xy, \qquad f_y = x^2 + e^y, \qquad \nabla f = (2xy, x^2 + e^y),$$

and

$$\nabla^2 f(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & e^y \end{pmatrix}.$$

Proposition 5.3 (Multivariable chain rule)**.** If $x(t) \in \mathbb{R}^d$ and f are differentiable, then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x(t)) = \nabla f(x(t)) \cdot x'(t).$$

For an option value $V = V(t, S)$ and a differentiable price path $S(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} V(t, S(t)) = V_t(t, S(t)) + V_S(t, S(t)) S'(t).$$

V_S measures sensitivity to the underlying price, V_t sensitivity to calendar time, and V_{SS} curvature with respect to the underlying.

Exercise — apply the chain rule

For $V(t, s) = e^{-rt} s^2$ and a differentiable path $S(t)$, compute $\frac{\mathrm{d}}{\mathrm{d}t} V(t, S(t))$.

Solution

Differentiate first with respect to each argument while holding the other one fixed:

$$V_t(t, s) = -r e^{-rt} s^2, \qquad V_s(t, s) = 2 e^{-rt} s.$$

The multivariable chain rule along the path $t \mapsto (t, S(t))$ gives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} V(t, S(t)) &= V_t(t, S(t)) + V_s(t, S(t)) S'(t) \\ &= -r e^{-rt} S(t)^2 + 2 e^{-rt} S(t) S'(t) \\ &= e^{-rt} \big(-r S(t)^2 + 2 S(t) S'(t) \big). \end{aligned}$$